

Replicating the Finned-Tube Heater CFD in ANSYS Fluent

Air-side heat-transfer validation of an eVinci-class heat-pipe reactor heater — porting the OpenFOAM v2 case to ANSYS

Combined Microreactor Project | Phase 1 Derating | CFD replication guide

For an ANSYS licence supporting ~80 M cells

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1. Purpose and scope

This guide contains everything needed to reproduce, in **ANSYS Fluent**, the air-side heat-transfer validation that was originally performed in OpenFOAM (`rhoSimpleFoam`, $k-\omega$ SST). The simulated object is a single **finned-tube unit cell** of the reactor heater: cross-flow air being heated by a hot, isothermal (condensing-sodium heat-pipe) tube wall. The deliverable of the CFD is one dimensionless number — the **Nusselt number** — validated against a published finned-tube correlation.

Target to reproduce: $Nu \approx 45.9$ at $Re = 8368$, within $\pm 20\%$ of the Briggs–Young correlation ($Nu_{BY} = 54.1$, i.e. a deviation of -15.2%). Reaching this certifies the air-side heat-transfer model.

What the ~80 M-cell licence unlocks (beyond the original single-cell, single-point scope): a formal grid-convergence study, a full multi-row staggered bank (removing the fin-clipping limitation), conjugate heat transfer in the fins, and a multi-point Reynolds sweep. These optional extensions are in §11.

2. Reference results to match

All values below are measured or derived from the converged OpenFOAM solution and are the acceptance targets for the ANSYS run.

Quantity	Symbol	Value	Notes
Wall heat rate (tube patch)	Q	256.3 W	surface integral of wall heat flux
Inlet–outlet pressure drop	Δp	148.4 Pa	area-averaged
Air outlet temperature	T_{out}	808.6 K	mass-weighted
Log-mean temperature diff.	ΔT_{lm}	257.1 K	driving ΔT (see §8)
Heat-transfer coefficient	h	$\sim 110 \text{ W m}^{-2} \text{ K}^{-1}$	$q''/\Delta T_{lm}$
Nusselt number	Nu	45.9	$h d_o/k$; the headline
Reynolds number	Re	8368	min-flow mass flux, tube OD
Friction factor	f	1.69	informational only (see §9)
Near-wall resolution	y^+	mean 0.24 / max 4.6	tube+fin walls
Mesh size (original)	—	3.29 M cells	baseline; refine with 80 M budget

3. Geometry

3.1 Reference finned tube (staggered bank)

Parameter	Symbol	Value
Tube outer diameter	d_o	25.4 mm (= 1 in)
Fin height	—	12.0 mm
Fin thickness	t	0.5 mm
Fin pitch (centre-to-centre)	p_f	4.0 mm
Inter-fin gap	$s = p_f - t$	3.5 mm
Transverse pitch	$S_T = 2.0 d_o$	50.8 mm
Longitudinal pitch	$S_L = 1.75 d_o$	44.45 mm
Fin material conductivity	k_{fin}	$25 \text{ W m}^{-1} \text{ K}^{-1}$ (high-temp alloy)

Provenance note. The eVinci heat-exchanger geometry is not public; these are representative standard finned-tube values (1-inch tube, conventional pitch ratios). See references [8, 9].

3.2 Computational domain (the “unit cell”)

One tube, spanning **three fin pitches** along its axis, with periodic/symmetry planes standing in for the infinite bank. Axis convention:

Axis	Extent	Physical meaning	Boundary type
x (transverse)	$\pm S_T/2 = \pm 25.4$ mm	tube-to-tube	translational periodic
y (streamwise)	$\pm S_L/2 = \pm 22.225$ mm	flow direction	inlet / outlet
z (tube axis, spanwise)	$0 \rightarrow 12$ mm ($3p_f$)	along the tube	symmetry (mid-fin-gap planes)

The fluid domain is the box **minus** the solid tube-plus-fins. Tube axis lies along z at $(x, y) = (0, 0)$; fins sit at $z = 2, 6, 10$ mm so the $z = 0$ and $z = 12$ mm faces fall on symmetry planes.

3.3 Build options in ANSYS

- **SpaceClaim / DesignModeler:** create the box, create the finned tube (cylinder + three annular fin discs), Boolean-subtract the solid from the box to get the fluid domain. Name the faces (**inlet**, **outlet**, **tube**, **periodic_x_pos/neg**, **sym_z_pos/neg**).
- **Or import the existing STL:** `phase1_derating/cfd/heater_unitcell/constant/triSurface/tube.stl` is the exact finned-tube surface; wrap/subtract it inside the box.

Fin-clipping caveat (inherited). The fin outer radius (24.7 mm) exceeds $S_L/2$ (22.225 mm), so in a single-tube box the fins are clipped at the inlet/outlet planes. This is inherent to one-tube domains and biases friction, not heat transfer. With 80 M cells you can instead build a **multi-row bank** (§11) to remove it.

4. Mesh

4.1 Strategy

- **Recommended:** Fluent Meshing **poly-hexcore** (Mosaic) with **boundary-layer inflation** on the tube walls, or ANSYS Meshing with sweep + inflation.
- **Thin fins (0.5 mm):** enforce local face sizing so ≥ 4 cells span the fin thickness and wrap inflation around the fin tips/edges.
- **Near-wall target:** $y^+ \leq 1$ on tube+fin walls (the original achieved mean 0.24). $k-\omega$ SST needs a wall-resolved mesh here — do **not** use wall functions.

4.2 First-cell height (inflation)

Flat-plate estimate at the min-flow velocity ($U_{max} \approx 13.7$ m/s, $L = d_o$, film properties): $C_f \approx 0.058 Re^{-0.2} \approx 9.5 \times 10^{-3}$, $\tau_w \approx 0.84$ Pa, $u_\tau \approx 0.94$ m/s, giving for $y^+ = 1$:

$$y_1 = \frac{y^+ \mu}{\rho u_\tau} \approx \frac{(1)(3.9 \times 10^{-5})}{(0.94)(0.94)} \approx 44 \mu\text{m}.$$

Recommendation: first-layer height **20–30 μm** , growth ratio **1.15–1.2**, **12–15 layers**; then *verify* the y^+ contour on **tube** after the solve and adjust (fin leading edges run higher). The original OpenFOAM first cell was $\sim 24 \mu\text{m}$ ($\Rightarrow y^+$ mean 0.24).

4.3 Grid-convergence study (use the 80 M budget)

Run **three systematically refined meshes** (e.g. ~ 8 M / ~ 25 M / ~ 80 M cells, refinement ratio ~ 1.5 per direction), report Nu on each, and compute the **Grid Convergence Index (GCI)** [10]. This replaces the single-mesh adequacy argument of the original (a documented limitation) with a formal mesh-independence result.

5. Physics and solver setup (Fluent)

Setting	Value	OpenFOAM equivalent
Solver	Pressure-based, Coupled scheme, Steady On	rhoSimpleFoam (compressible SIMPLE)
Energy equation	On	—
Viscous model	k - ω SST , low-Re / wall-resolved	kOmegaSST (RAS)
Operating pressure	200 000 Pa (2 bar, heater side)	p field $\approx 2\text{e}5$
Gravity	Off	—

5.1 Material — air (temperature-dependent)

Property	Method in Fluent	Value / coefficients
Density	ideal-gas	$\rho = p/(RT)$, $R = 287 \text{ J kg}^{-1} \text{ K}^{-1}$
Viscosity	Sutherland (two-coefficient)	$\mu = \frac{1.458 \times 10^{-6} T^{1.5}}{T + 110.4}$
Specific heat c_p	piecewise-poly (JANAF) or constant	$\sim 1080 \text{ J kg}^{-1} \text{ K}^{-1}$ at film temp
Thermal conductivity k	set to hold $Pr = 0.69$	$k = \mu c_p / Pr \approx 0.061 \text{ W m}^{-1} \text{ K}^{-1}$ at film

Consistency note. The validation fixes the air Prandtl number at $Pr = 0.69$ (high-temperature air) and derives $k = \mu c_p / Pr$. Reproduce this so that $Nu = hd_o/k$ is comparable with the Briggs–Young correlation (which is evaluated at the same Pr). If you instead use kinetic-theory k , report the actual Pr used.

6. Boundary conditions

Face	Fluent BC	Settings
inlet	velocity-inlet	$U = 6.3623 \text{ m s}^{-1}$ (normal, $+y$); $T = 740 \text{ K}$; turb. intensity $\sim 5\%$, viscosity ratio ~ 10
outlet	pressure-outlet	gauge 0 Pa (operating 200 kPa); backflow $T = 740 \text{ K}$, backflow turb. as inlet
tube (+ fins)	wall	no-slip; fixed temperature $T_w = 1033 \text{ K}$ (condensing-Na heat-pipe wall)
periodic_x_*	periodic	translational , offset $S_T = 50.8$ mm
sym_z_*	symmetry	mid-fin-gap planes

The isothermal wall represents the sodium heat-pipe condenser, which is genuinely near-isothermal by phase change. If you enable conjugate heat transfer (§11), instead set the fin solid with $k_{fin} = 25 \text{ W m}^{-1} \text{ K}^{-1}$ and apply 1033 K at the tube bore.

7. Solution controls and convergence

- **Spatial discretization:** second-order upwind for momentum, energy, turbulence; PRESTO! or second-order for pressure.
- **Coupled controls:** pseudo-transient or a Courant number ramp; keep density/energy under-relaxation moderate early (the original used heavy under-relaxation: p 0.3, ρ 0.05).
- **Initialization:** hybrid initialization, or standard from inlet values.

- **Convergence:** scaled residuals $< 10^{-4}$ (target 10^{-5} – 10^{-6}), **and** monitor to steadiness:
 - (i) area-weighted **wall heat flux on tube**, (ii) mass-weighted **outlet temperature**, (iii) mass imbalance inlet vs outlet $< 0.1\%$. The original converged in 464 iterations.
- **Temperature limiter (optional):** clamp T to a physical band (e.g. 300–1200 K) to survive early transients.

8. Post-processing — computing Nu , Re , f

Extract three integrals from the converged solution (Reports → Surface Integrals / Fluxes):

1. Q = integral of *Total Surface Heat Flux* over **tube** → target 256 W.
2. T_{out} = mass-weighted average of temperature on **outlet**.
3. $\dot{m}$ = mass flow rate at **inlet**; Δp = area-avg $p(\text{inlet}) - \text{area-avg } p(\text{outlet})$.

Then compute (property model: $R = 287$, $c_p = 1080$, $Pr = 0.69$, Sutherland μ):

$$q'' = \frac{Q}{A_{wetted}}, \quad \Delta T_{lm} = \frac{\Delta T_{in} - \Delta T_{out}}{\ln(\Delta T_{in}/\Delta T_{out})}, \quad \Delta T_{in} = T_w - T_{in}, \quad \Delta T_{out} = T_w - T_{out}.$$

$$h = \frac{q''}{\Delta T_{lm}}, \quad T_{film} = \frac{1}{2} \left(T_w + \frac{T_{in} + T_{out}}{2} \right), \quad Nu = \frac{h d_o}{k(T_{film})}.$$

$$Re = \frac{G_{max} d_o}{\mu(T_{film})}, \quad G_{max} = \frac{\dot{m}}{A_{min}}, \quad A_{min} = 3[(S_T - d_o)p_f - 2(\text{fin height})t] = 2.69 \times 10^{-4} \text{ m}^2.$$

$$f = \frac{\Delta p}{\frac{1}{2} \rho_{in} U_{max}^2 N_{rows}}, \quad U_{max} = \frac{\dot{m}}{\rho_{in} A_{min}}.$$

Critical convention — use LMTD, not wall-minus-inlet. The air warms across the tube, so the driving ΔT is the log-mean. Using the naïve $T_w - T_{in}$ gives $Nu = 40.3$ (−25.5%, **fails** validation); LMTD gives $Nu = 45.9$ (−15.2%, **passes**). Briggs–Young is itself LMTD-based, so LMTD is the like-for-like choice.

9. Validation correlations

Evaluate at the CFD-derived Re and Pr .

Briggs–Young (1963) — heat transfer [1]:

$$Nu_{BY} = 0.134 Re^{0.681} Pr^{1/3} \left(\frac{s}{H_{fin}} \right)^{0.2} \left(\frac{s}{t} \right)^{0.1134}.$$

With $s = 3.5$ mm, $H_{fin} = 12$ mm, $t = 0.5$ mm ($s/H_{fin} = 0.292$, $s/t = 7.0$) at $Re = 8368$, $Pr = 0.69$: $Nu_{BY} = 54.1$. **Acceptance: CFD within $\pm 20\%$ → validated.**

Robinson–Briggs (1966) — friction [2]:

$$f_{RB} = 9.465 Re^{-0.316} \left(\frac{S_T}{d_o} \right)^{-0.927} = 0.287 \text{ at } Re = 8368, \quad S_T/d_o = 2.0.$$

Friction caveat. In the single-tube box the CFD Δp spans the whole domain (plenum + wake), while Robinson–Briggs describes only the per-row bundle loss; the two are not directly comparable, so friction is reported but **not used**. A multi-row bank (§11) makes f meaningful.

10. Expected outcome and checks

Check	Expected	Meaning
Nu (LMTD)	44–48 (orig. 45.9)	within $\pm 20\%$ of $Nu_{BY} = 54.1 \rightarrow$ validated
y^+ on tube	mean < 1	wall-resolved; $k-\omega$ SST valid
Q	~ 256 W (single cell)	energy into the air
Mass imbalance	$< 0.1\%$	converged
GCI (3 meshes)	$< \text{few } \%$	mesh-independent (new, vs original)

If Nu lands materially outside $\pm 20\%$, first check: (a) LMTD used, not $T_w - T_{in}$; (b) $Pr = 0.69$ and k consistent; (c) $y^+ \leq 1$; (d) Re computed on the **min-flow** section, not inlet velocity.

11. Extensions enabled by the 80 M-cell budget

1. **Grid-convergence (GCI) study** — 3 meshes; converts the original single-mesh adequacy claim into a formal mesh-independence proof.
2. **Multi-row staggered bank** (e.g. 4–6 rows $\times$ several columns with true periodicity) — removes fin clipping and makes the **friction factor** validate against Robinson–Briggs.
3. **Conjugate heat transfer (CHT)** — model the fin as solid with $k_{fin} = 25 \text{ W m}^{-1} \text{ K}^{-1}$ and 1033 K at the tube bore, obtaining the **fin efficiency** ($\eta_{fin} \approx 0.53$) directly rather than analytically.
4. **Reynolds sweep** (6 points, e.g. $Re = 4000\text{--}14000$) — a fully CFD-derived $Nu(Re)$ law, replacing the correlation-anchored curve.

12. OpenFOAM $\rightarrow$ ANSYS quick map (appendix)

OpenFOAM	ANSYS Fluent
rhoSimpleFoam (steady compressible SIMPLE)	Pressure-based Coupled, Steady
kOmegaSST	$k-\omega$ SST
thermophysicalProperties (perfectGas, Sutherland, JANAF)	ideal-gas density, Sutherland viscosity, piecewise c_p
fixedValue U inlet	velocity-inlet
inletOutlet / pressure outlet	pressure-outlet
fixedValue T=1033 on tube	wall, fixed temperature 1033 K
cyclic (x)	translational periodic
symmetryPlane (z)	symmetry
wallHeatFlux, yPlus function objects	Surface Integrals / y^+ contour
snappyHexMesh + 8 layers	Fluent Meshing poly-hexcore + inflation

References

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